

PRACTICAL 9

AIM

Write the test cases for the identified module.

Objectives

1. Identify the module to be tested.
2. Write comprehensive test cases to validate the module's functionality.
3. Ensure the module meets the system's requirements and performance standards.

Prerequisites

1. Basic understanding of software testing concepts.
2. Knowledge of the PEFR Health Management System's architecture and modules.
3. Familiarity with test case writing techniques.

Tools and Resources

1. Test case management tool (e.g., TestRail, Zephyr, or Excel).
2. Access to the PEFR Health Management System module.
3. Sample PEFR data for testing.
4. Documentation of the module's requirements and specifications.

Lab Tasks

Step 1: Identify the Module

- Select the module to be tested (e.g., PEFR Data Input Module, AI Prediction Module, User Notification Module, etc.).
- Review the module's requirements and specifications.

Step 2: Define Test Scenarios

- Identify key functionalities of the module.
- Define test scenarios covering normal, edge, and error cases.



Step 3: Write Test Cases

- Write detailed test cases for each scenario.
- Include the following components in each test case:
 1. **Test Case ID:** Unique identifier for the test case.
 2. **Description:** Brief description of the test case.
 3. **Preconditions:** Conditions required before executing the test.
 4. **Test Steps:** Step-by-step instructions to execute the test.
 5. **Expected Result:** Expected outcome of the test.
 6. **Actual Result:** To be filled during test execution.
 7. **Status:** Pass/Fail.

Step 4: Review and Finalize

- Review the test cases for completeness and accuracy.
- Finalize the test cases for execution.

Example Test Cases

Module: PEFR Data Input Module

Test Case 1: Valid PEFR Data Input

- **Test Case ID:** TC_PEFR_001
- **Description:** Verify that the system accepts valid PEFR data.
- **Preconditions:** User is logged in, and the system is ready to accept input.
 - **Test Steps:**
 1. Enter a valid PEFR value (e.g., 400 L/min).
 2. Click the "Submit" button.
- **Expected Result:** The system accepts the input and displays a success message.
- **Actual Result:** [To be filled during execution]
- **Status:** [To be filled during execution]

Test Case 2: Invalid PEFR Data Input

- **Test Case ID:** TC_PEFR_002
- **Description:** Verify that the system rejects invalid PEFR data.

- **Preconditions:** User is logged in, and the system is ready to accept input.
 - **Test Steps:**
 1. Enter an invalid PEFR value (e.g., -100 L/min).
 2. Click the "Submit" button.
- **Expected Result:** The system displays an error message and rejects the input.
- **Actual Result:** [To be filled during execution]
- **Status:** [To be filled during execution]

Test Case 3: PEFR Data Input with Missing Values

- **Test Case ID:** TC_PEFR_003
- **Description:** Verify that the system handles missing PEFR data.
- **Preconditions:** User is logged in, and the system is ready to accept input.
 - **Test Steps:**
 1. Leave the PEFR value field empty.
 2. Click the "Submit" button.
- **Expected Result:** The system displays an error message prompting the user to enter a value.
- **Actual Result:** [To be filled during execution]
- **Status:** [To be filled during execution]

Module: AI Prediction Module

Test Case 4: Accurate PEFR Trend Prediction

- **Test Case ID:** TC_AI_001.
- **Description:** Verify that the AI module accurately predicts PEFR trends.
- **Preconditions:** Historical PEFR data is available in the system.
 - **Test Steps:**
 1. Provide a set of historical PEFR data.
 2. Run the AI prediction algorithm.
- **Expected Result:** The system generates accurate trend predictions based on the input data.
- **Actual Result:** [To be filled during execution]

- **Status:** [To be filled during execution]

Test Case 5: Prediction with Insufficient Data

- **Test Case ID:** TC_AI_002
- **Description:** Verify that the AI module handles insufficient data gracefully.
- **Preconditions:** Limited or no historical PEFR data is available.
 - **Test Steps:**
 1. Provide insufficient or no historical PEFR data.
 2. Run the AI prediction algorithm.
- **Expected Result:** The system displays a message indicating insufficient data for prediction.
- **Actual Result:** [To be filled during execution]
- **Status:** [To be filled during execution]

Deliverables

1. A document containing all written test cases.
2. Screenshots or logs of test execution (if applicable).
3. A summary report of the test case review and finalization process.

Conclusion

This lab manual provides a structured approach to writing test cases for the AI-enhanced PEFR Health Management System. By following these steps, you can ensure that the identified module is thoroughly tested and meets the system's requirements. Proper testing will enhance the reliability and performance of the system, ultimately improving respiratory health management for users.